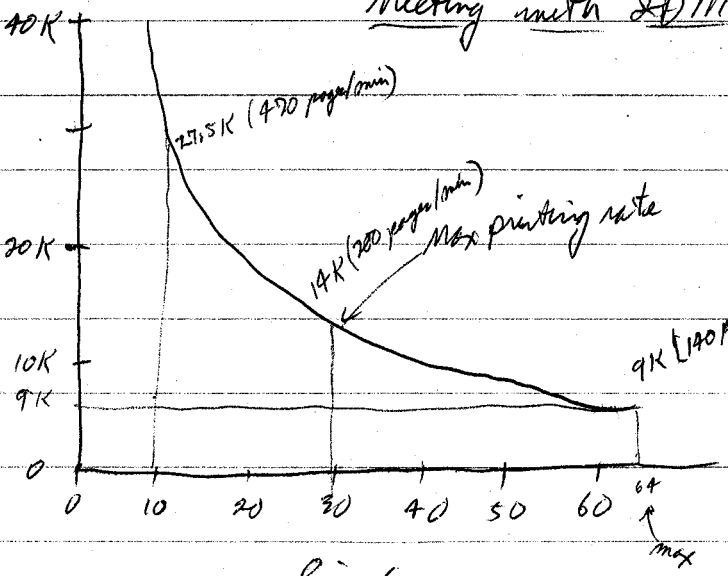
IBM Men
1. Bob Evans
2. Rogers
3. Larry
4. —

(1st meeting with Bob Evans)

Nov 9, 1956

Meeting with IBM on electronic printer

Speed:



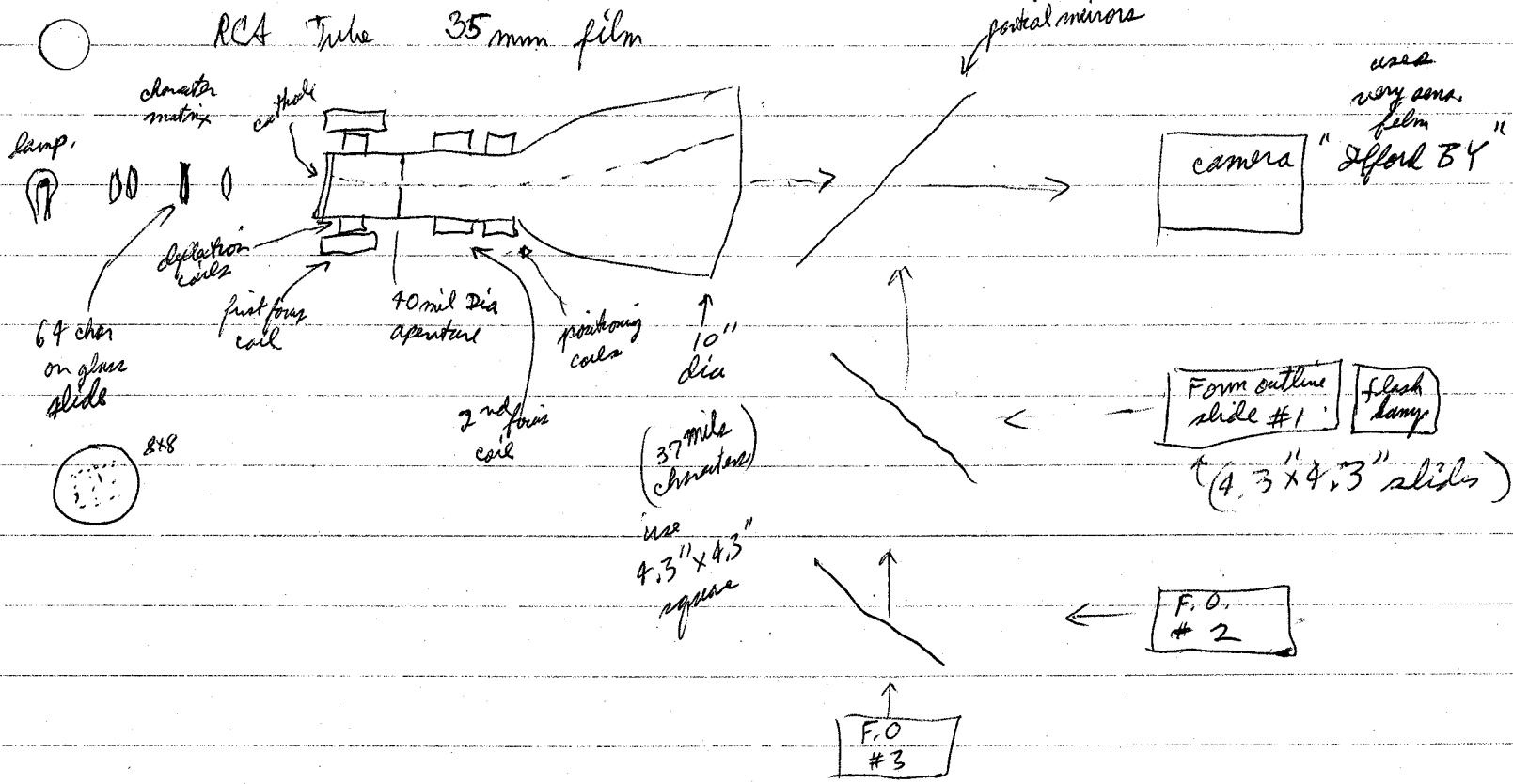
lines/page
(assume 128 char per line)

assumptions
1. 128 char. per line
60 μs per char (except for 1st)
2. 20 ms = frame change time is used for printing (out of 85-100 ms)

64 lines per page
140 page/min

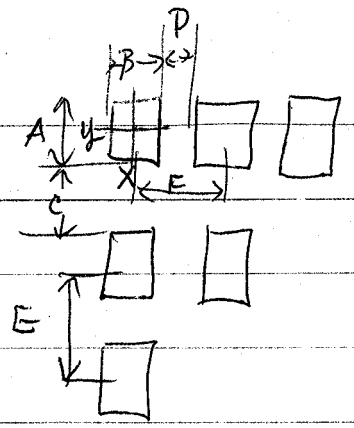
(100 ms to advance film)

RCA Tube 35 mm film



13 x 13 blow-up planned for 35 mm film ? can be smaller

want to use square if possible.



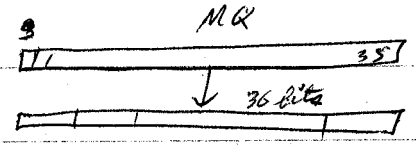
character dimensions (coord of character

	microfilm	Page (13X13)
A	.144m	.100 in
B	.108	.075
C	.144	.100
D	.036	.025
E	.288	.200
F	.288	.144 .100

5 lines/in

characters go on one at a time (200 μs decay time phosphor)

dot .003" accuracy, on tube face.



has buffer reg. 36 bits from MQ

buffer sends out (6 bits) at a time to printer

5 1 2 3 4 5	control characters
0 0 0 0 0 0	no op
0 0 1 0 0 0	print, starting at specified point
0 1 0 0 0 0	" " " current point
0 1 1 0 0 0	Plot selected character
1 0 0 0 0 0	Print form outline 1
1 0 1 0 0 0	" 2
1 1 0 0 0 0	" 3
1 1 1 0 0 0	advance Film frame
0 0 0 1 0 0	generate x axis
0 0 0 1 0 0	" y axis

3-5	8-17	18-23	26-35
control	$(X)_8$	char.	$(Y)_8$

word use for plotting

char 1	char 2	char 3	char 4	char 5	char 6

word use for printing
six characters per wd.

printing same as plotting where
X & y are understood.

at end of 128 jumps back
to beginning of next line space
down one - at end of page get
end of record skip.
to print at different position

Future possibilities:

camera → Film Processor → views

dry process

presently only 2 sec.

I. Print 14 characters

(a) starting at $X=(60)_8$ $y=(150)_8$

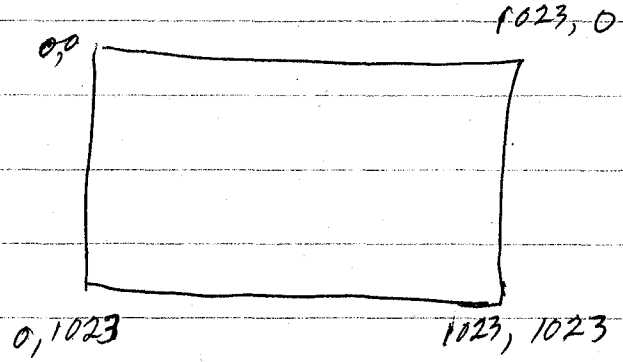
(b) continuing from current diff. position

WRS Microfilm Br.
LXA L(4) A
1,0 copy L(Data wd 1)+A, A
TIX 1,0, A, 1

	Data Wd.	5-5	6-11	12-17	18-23	24-29	30-35
(a)	1	(Print 001000)	0	60	—	01	50
	2	A	B	C	D	E	F
	3	G	H	I	J	K	L
	4	M	N	(Signal 111111)			

		Print current address 5-5		(no offset)			
(b)	1	(610 000)		—	—	—	—
	2	A	B	C	D	E	F
	3	G	H	I	J	K	L
	4	M	N	(Signal 111111)			

III, The origin of the graph to be plotted is (524, 500)
plot (a) 2, 6 (b) −10, 20 (c) 34, −61 (d) 80, −160



(a) $524 + 2 = (526)_{10} = (1016)_8$ (x)

(b) $500 - 6 = (494)_{10} = (0756)_8$ (y)

(b) $524 - 10 = (514)_{10} = (1002)_8$ x
$500 - 20 = (0740)_8$ y

(c) $(1056)_8$ x
$(1061)_8$ y

(d) 0674 x
1224 y

(6-7) not used (24,25) not used

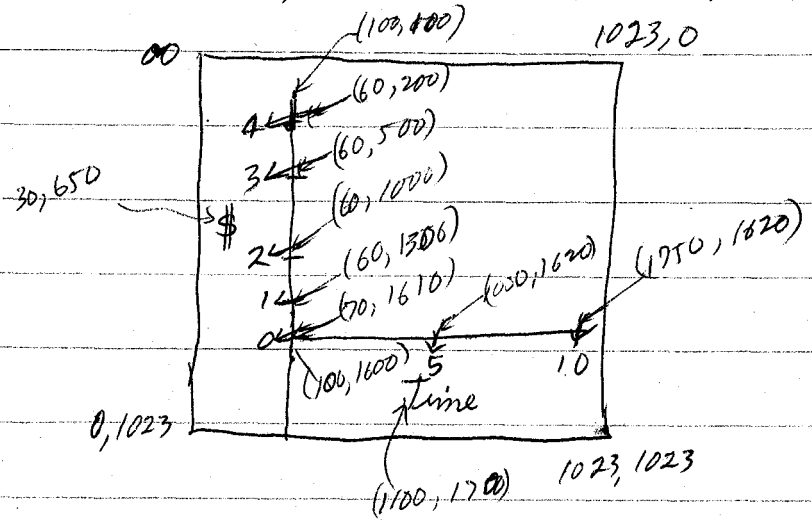
Data Word:	5-5	6-17	18-23	24-35
	Plot (011000)		+ character 010000	
1		1016		0756
2	00	1002	—	0740
3	00	1056	—	1061
4	00	0674	—	1224

These mean continue previous operation (i.e. print + plot)

For buffered output:

WRS M.P.
RLA (word count, FWA) Read Load channel A

IV. Show the control words & data required to create 2 axes, label them, superimpose film frame and advance



Word #	5-5	6-11	12-17	18-23	24-29	30-35	
1	(111000) adv. film	—	—	—	—	—	
2	(000100) generate X axis	01	00	(dash 100000)	16	00	(dashes start at 100,1600)
3,	(001100) gen. Y axis	01	00	(vertical dash)	01	00	(down from 100,100)
4.	(100000) Print Form 1	—	—	—	—	—	
5.	(011000 plot	00	60	(4)	02	00	mark off y
6.	00	00	60	(3)	05	00	
7.	00	00	60	(2)	10	00	
8.	00	00	60	(1)	13	00	
9,	00	00	30	(#)	06	50	label y
10.	00	00	70	(0)	16	10	origin
11.	00	10	00	(5)	16	20	
12.	Print (00100)	17	00	—	16	20	
13.	1	0	(signal 11111)	—	—	—	
14.	Print (00100)	11	00	—	17	00	label X
15.	T	1	M	E	(signal 1111)	—	